

Chapter 4 | Types of Chemical Reactions and Solution Stoichiometry

4.1 Water, the Common Solvent

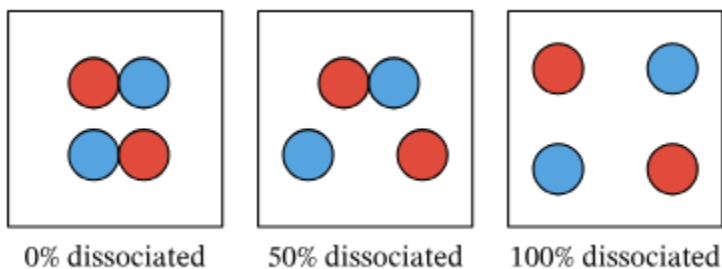
- **Properties of Water:**
 - Polarity and hydrogen bonding.
 - High specific heat, surface tension, and solvent capabilities.
- **Diagram of Water Molecule:** (Leave space for drawing)

Key Concepts:

- _____
 - _____
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4.2 The Nature of Aqueous Solutions: Strong and Weak Electrolytes

- **Definition of Electrolytes:**
 - **Strong Electrolytes:** Completely dissociate in water (e.g., NaCl, HCl).
 - **Weak Electrolytes:** Partially dissociate in water (e.g., CH_3COOH).
- **Examples and Diagrams:**
 - Strong Electrolytes: _____
 - Weak Electrolytes: _____
- **Diagram of Dissociation:** (Leave space for drawing)



CHEMICAL CONNECTIONS: Arrhenius: A Man with Solutions

- **Arrhenius's Contributions:**
 - Definition of acids and bases.

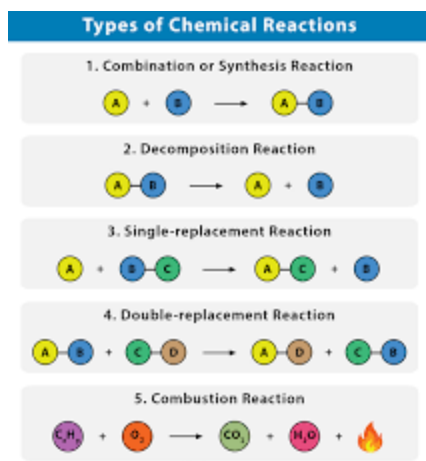
- Importance in understanding electrolytes.
 - **Key Points:**
 - Arrhenius acids donate H^+ to a solution ($HCl \rightarrow H^+ + Cl^-$)
 - Arrhenius bases donate OH^- to a solution ($NaOH \rightarrow Na^+ + OH^-$)
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4.3 The Composition of Solutions

- **Solution Composition Terms:**
 - **Solvent:** The substance in greater amount.
 - **Solute:** The substance dissolved in the solvent.
 - **Concentration Units:**
 - **Molarity (M):** Moles of solute per liter of solution.
 - **Mass Percent:** Mass of solute per mass of solution times 100.
 - **Example Calculations:**
 - Molarity: _____
 - Mass Percent: _____
 - **Practice Problems:**
 - Calculate the molarity of a solution containing 5 grams of NaCl in 250 mL of water: _____
 - Determine the mass percent of a solution with 10 grams of solute in 100 grams of solution: _____
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4.4 Types of Chemical Reactions

- **Classification of Reactions:**
 - **Synthesis:** $A + B \rightarrow AB$
 - **Decomposition:** $AB \rightarrow A + B$
 - **Single Displacement:** $A + BC \rightarrow AC + B$
 - **Double Displacement:** $AB + CD \rightarrow AD + CB$
 - **Combustion:** $\text{Hydrocarbon} + O_2 \rightarrow CO_2 + H_2O$
- **Examples:**
 - _____
 - _____
- **Diagram of Reaction Types:** (Leave space for drawing)



4.6 Describing Reactions in Solution

- **Molecular, Complete Ionic, and Net Ionic Equations:**
 - **Molecular Equation:** Shows complete formulas of reactants and products.
 - **Complete Ionic Equation:** Shows all ions present in the reaction.
 - **Net Ionic Equation:** Shows only the species that actually change during the reaction.
- **Example:**
 - Molecular: $\text{AgNO}_3(\text{aq}) + \text{NaCl}(\text{aq}) \rightarrow \text{AgCl}(\text{s}) + \text{NaNO}_3(\text{aq})$
 - Complete Ionic: $\text{Ag}^+(\text{aq}) + \text{NO}_3^-(\text{aq}) + \text{Na}^+(\text{aq}) + \text{Cl}^-(\text{aq}) \rightarrow \text{AgCl}(\text{s}) + \text{Na}^+(\text{aq}) + \text{NO}_3^-(\text{aq})$
 - Net Ionic: $\text{Ag}^+(\text{aq}) + \text{Cl}^-(\text{aq}) \rightarrow \text{AgCl}(\text{s})$
- **Practice Problems:**
 - Write the net ionic equation for the reaction between K_2SO_4 and $\text{Pb}(\text{NO}_3)_2$:

4.7 Stoichiometry of Precipitation Reactions

- **Calculations Involving Precipitation:**
 - Determine the amount of precipitate formed.
 - Calculate the concentration of ions remaining in solution.
- **Example Calculation:**
 - If 50 mL of 0.1 M AgNO_3 is mixed with 50 mL of 0.1 M NaCl , how much AgCl is formed?
 - Steps: _____
 - _____
- **Practice Problems:**

- Determine the mass of CaCO_3 formed when 100 mL of 0.2 M CaCl_2 is mixed with 100 mL of 0.2 M Na_2CO_3 : _____
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4.8 Acid–Base Reactions

- **Definition and Examples:**
 - **Acid:** Proton donor (e.g., HCl , H_2SO_4).
 - **Base:** Proton acceptor (e.g., NaOH , NH_3).
 - **Neutralization Reaction:**
 - $\text{Acid} + \text{Base} \rightarrow \text{Salt} + \text{Water}$
 - Example: $\text{HCl} + \text{NaOH} \rightarrow \text{NaCl} + \text{H}_2\text{O}$
 - **pH Scale:**
 - _____
 - _____
 - **Diagram of pH Scale:** (Leave space for drawing)
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4.9 Oxidation–Reduction Reactions

- **Definition and Examples:**
 - **Oxidation:** Loss of electrons.
 - **Reduction:** Gain of electrons.
 - **Oxidizing and Reducing Agents:**
 - **Oxidizing Agent:** Gains electrons and is reduced.
 - **Reducing Agent:** Loses electrons and is oxidized.
 - **Example Reaction:**
 - $\text{Zn} + \text{Cu}^{2+} \rightarrow \text{Zn}^{2+} + \text{Cu}$
 - **Diagram of Redox Reaction:** (Leave space for drawing)
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4.10 Balancing Oxidation–Reduction Equations

- **Steps for Balancing Redox Equations:**
 - Separate into half-reactions.
 - Balance atoms and charges.
 - Combine half-reactions.
- **Example:**
 - Balance $\text{MnO}_4^- + \text{Fe}^{2+} \rightarrow \text{Mn}^{2+} + \text{Fe}^{3+}$ in acidic solution.
 - Steps: _____
 - _____
- **Practice Problems:**

- Balance the redox equation: $\text{Cr}_2\text{O}_7^{2-} + \text{Cl}^- \rightarrow \text{Cr}^{3+} + \text{Cl}_2$ in acidic solution:
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4.11 Simple Oxidation–Reduction Titrations

- **Purpose and Procedure:**
 - Determine the concentration of an unknown solution using a redox reaction.
 - Example: Titration of Fe^{2+} with KMnO_4 .
 - **Example Calculation:**
 - Calculate the concentration of Fe^{2+} in a solution if 25.0 mL of 0.02 M KMnO_4 is required to titrate 50.0 mL of Fe^{2+} solution.
 - Steps: _____
 - _____
 - **Practice Problems:**
 - Determine the concentration of H_2O_2 if 30.0 mL of 0.1 M KMnO_4 is required to titrate 100 mL of H_2O_2 solution: _____
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For Review

- Summarize the key concepts from the chapter.
 - List any questions or areas that need further clarification.
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Key Terms

- **Electrolyte:** _____
- **Precipitate:** _____
- **Redox Reaction:** _____
- **Molarity:** _____